

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder with over 4 years of experience as a Software Developer and Data Engineer, proficient in building scalable microservices and distributed data pipelines. Leveraging expertise in Java, Python, Spark, SQL, AWS, and Azure, alongside fullstack development capabilities, to drive decision-making and deliver robust solutions in a Senior Software Engineer role. Experienced in CI/CD automation and graduate-level research on large-scale data platforms and privacy-preserving AI applications.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered and optimized high-performance REST and SOAP APIs using Java 8/11 and Spring Boot, successfully handling over 50,000 daily requests with 99.99 percent SLA uptime.
- Built and automated robust CI/CD pipelines leveraging Jenkins and Azure DevOps, facilitating continuous delivery across Azure Cloud environments.
- Modernized legacy systems by migrating to Log4j 2.x, enhancing monitoring and error handling mechanisms, reducing issue resolution time by 20 percent.
- Spearheaded an EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, employing Git and JUnit for comprehensive testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for key automotive clients.
- Boosted system performance by 90 percent through meticulous SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented efficient CI/CD pipelines using Jenkins and GitLab, incorporating automated testing and stringent quality gates for enhanced code integrity.
- Mentored junior developers, fostering best practices through detailed code reviews, debugging sessions, and guidance on clean coding principles.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems, achieving over 99 percent uptime through proactive monitoring and continuous performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening overall data integrity.
- Implemented comprehensive unit testing strategies using JUnit4, successfully achieving 99 percent code coverage across critical modules.
- Supported release planning and sprint execution utilizing JIRA within dynamic Agile development teams.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, facilitating rapid insights.
- Developed an interactive Flask web application for dynamic data exploration with advanced filtering and visualization capabilities.
- Containerized ingestion, processing, and analytics components using Docker, ensuring reproducible deployments and simplified workflow management.
- Optimized Spark memory usage and partitioning strategies, resulting in a significant 40 percent reduction in overall job runtime.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight for scalable data solutions.

Privacy-Preserving AI Email Client (Capstone Project, T5, LoRA, Python) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform advanced email summarization and classification tasks.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under stringent privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment of AI models.

SKILLS

- Programming Languages: Java, Python, SQL, Kotlin, Javascript, Typescript.
- Frameworks & Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, React, Node.js.
- Web Services: REST APIs, SOAP APIs, GraphQL.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.
- Databases: MongoDB, PostgreSQL, Oracle Database, IBM DB2, RDS.

- Tools & Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform, Ui.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Workflow optimization, Experiment design, Decision-Making, Boosting algorithms, Fullstack development.